Radeon Motion Lab: An Evidence-Led Audit of Vision-Guided RGB-D Manipulation Robustness

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Abstract

Reliable manipulation evaluation must distinguish a policy that sees from a controller that merely receives simulator truth. We present Radeon Motion Lab, an auditable RGB-D manipulation system and perturbation protocol for a Franka Panda fruit pick-and-place task in Genesis on an AMD Radeon GPU through ROCm. The upstream scripted controller queries the target pose directly; our primary policy removes that Oracle input and estimates a grasp pose from RGB-D images, public camera calibration, known table geometry, a requested fruit class, and robot state. We freeze eleven causal conditions, vary one axis at a time, keep failures, and record perception error separately from task completion. The frozen baseline contains three YCB fruits, eleven conditions, and five seeds per condition. The ten runtime conditions complete 144/150 episodes, and a scene-rebuilt appearance matrix completes 15/15. Nine conditions complete 15/15; pose shift and low friction each complete 12/15 (Wilson 95% interval 0.548–0.930). All six failures are lemon execution failures despite evaluator-only XY perception errors below 0.70 mm. Failure-led sensor-only corrections convert retained seed-23 failures from plum no-candidate and lemon task failure into successful regression episodes, while a matched physical guard ablation decreases the physical-subset result from 27/30 to 20/30. A subsequent sensor-only controller follow-up adds finger proprioception, contact-triggered force branches, RGB-D lift checks, and bounded retries. Its final fixed lemon pose gate completes 4/5, so the candidate is not promoted. The contribution is therefore not a claim of universal manipulation; it is a reproducible method for locating the boundary between geometric perception, calibrated execution, and contact robustness on an AMD GPU.

1 Introduction

Manipulation demonstrations often compress several sources of information into one apparently successful rollout. In simulation, the most consequential shortcut is an action module that reads the object pose, bounding box, or instance label from the simulator. Such a controller can be a useful upper bound for inverse kinematics and dynamics, but it does not answer whether a robot can localize an object from sensors.

This paper studies that information boundary in a deliberately small system. Radeon Motion Lab picks a banana, lemon, or plum with a Franka Panda and places it in a calibrated bowl fixture. The policy uses a world-camera RGB-D frame and class-conditioned geometric rules. Its output is an estimated world-frame grasp center and jaw yaw, which are passed to a shared scripted IK executor. The evaluator may read simulator truth to score the initial localization and the final task state, but that truth is excluded from action selection.

The system is evaluated on an AMD Radeon environment using Genesis 1.3.1 and the `gs.amdgpu` backend. Rather than aggregate a single success number, we define orthogonal visual, depth, viewpoint, appearance, spatial, and physical perturbations. This follows a recent robotics trend toward testing closed-loop behavior under distribution shift and auditing the evidence on which actions depend, including geometry-aware viewpoint robustness[4, 5] and failure-oriented monitoring[7, 8, 9]. Our focus is complementary: a classical policy makes the information boundary and the causal intervention explicit enough to inspect frame by frame.

Our contributions are:

- a sensor-only RGB-D grasp estimator with calibrated back-projection, class-specific shape reasoning, and a documented exclusion of simulator entity state;
- an orthogonal perturbation matrix with seeded sampling, absolute per-episode dynamics reset, separate build-time appearance evaluation, and archived JSON/frame/video evidence;
- a failure-led analysis that separates localization error from task outcome, records contact and lift phases, and retains negative or partial controller ablations; and
- a measured AMD GPU benchmark receipt: 165 baseline cells, reproducible commands, environment metadata, checksums, and a public repository.

2 Related Work

Geometry-aware manipulation. Recent manipulation systems increasingly make camera geometry an explicit part of the representation. RayViT injects camera rays into visual imitation learning and reports improved robustness to camera perturbations[4]; OC-VLA++ trains cross-view action consistency to improve unseen-view generalization[5]. These learned approaches motivate our calibrated camera-view condition. We use an analytically back-projected point cloud rather than learning a representation, which makes the source of a localization error inspectable and provides a low-compute reference point for future learned baselines.

Generalization and failure auditing. BridgeVLA++ addresses data efficiency, memory, and distributional generalization in 3-D VLA manipulation[6]. GUARD probes the dependence of diffusion-VLA actions on multimodal evidence for test-time failure detection[7], while SAFECAST studies contrast-set training and calibration for failure probes under deployment shifts[8]. The closed-loop audit of task-vector editing likewise argues that target behavior and unrelated control must be evaluated together[9]. Our protocol shares this evaluation stance but exposes it at the system level: each policy input, perturbation draw, estimate, evaluator-only truth snapshot, and final outcome is retained. The resulting failure labels are deliberately coarse when the available telemetry cannot support a narrower causal claim.

Simulation and reproducibility. Genesis provides the differentiable, GPU-accelerated simulation substrate for robotics experiments[1], part of a broader move toward low-cost, high-fidelity simulators for robot learning[10]. The YCB object set supplies recognizable manipulation objects and a common naming convention[2]. Domain randomization is a standard route for exposing policies to appearance and state variation[3]; here, we use controlled perturbations as an evaluation instrument rather than claiming sim-to-real transfer.

3 System and Information Boundary

3.1 Task and runtime

The scene contains a Franka Panda[11], a table, three YCB fruit assets, a world camera, and a fixed calibrated bowl. Each episode requests one fruit and executes the sequence reach, descend, close, lift, transport, release, and retreat. The final success predicate is the upstream simulator's task check after the policy has completed its actions. The bowl is a known fixture in this version; it is not visually localized.

The observed AMD environment reports PyTorch 2.9.1 with ROCm 7.2.1, target `gfx1100`, the generic device name "AMD Radeon Graphics", and 47.98 GB of device memory as reported by Genesis. The simulator, rendering, and robot execution loop use `gs.amdgpu`. The detector uses classical OpenCV/NumPy operations; no learned-detector acceleration claim is made.

3.2 Policy inputs and forbidden state

The action path consumes only:

1. the world-camera RGB image and linear depth image;
2. public camera intrinsics and camera-to-world extrinsics;
3. robot state, known table height, and the requested fruit class; and
4. the configured pose of the fixed bowl fixture.

It does not consume fruit pose or quaternion, AABB, Genesis instance masks, ground-truth object IDs, or evaluator results. The evaluator snapshots the initial simulator state before action selection only to compute a localization metric; this snapshot is not passed to the policy. After execution it reads the final state to score success. The upstream simulator-state controller is therefore reported only as an Oracle IK upper bound, not as a visual baseline.

3.3 RGB-D geometric estimator

For image pixel (u, v) and linear depth d, calibrated back-projection is

$$\begin{aligned} \mathbf{p}_c &= d\left[\frac{u+0.5-c_x}{f_x}, \frac{v+0.5-c_y}{f_y}, 1\right]^\top, \\ \mathbf{p}_w &= T_{wc}\mathbf{p}_c. \end{aligned} \tag{1}$$

The policy first creates a class-conditioned HSV mask, applies small morphological opening and closing, and intersects the mask with points at least 10 mm above the known table and inside a reachable workspace. Connected components with fewer than 80 valid RGB-D points are rejected.

Banana and lemon share a yellow mask and are separated by component geometry. The banana selects the component with the largest robust major span and uses the first principal axis of its XY point cloud as jaw yaw. The lemon selects a compact component. The plum uses a dark-magenta mask. For nearly round lemon and plum targets, PCA yaw is unstable, so yaw is fixed for the top-down grasp. The estimate includes the selected component's area, robust major/minor spans, pixel center, position, yaw, and a diagnostic confidence value.

The visible depth points of a round object lie on its camera-facing surface. For lemon and plum, the policy instead casts the component-center pixel ray through the public calibration and intersects it with a class-calibrated center height above the table. The heights are 0.0265085 m for lemon and 0.0265200 m for plum. These are task calibration constants, not runtime object queries.

3.4 Shared executor and evaluation boundary

The estimated position sets the pre-grasp and vertical descent target. A top-down quaternion is formed from the estimated yaw, and the existing IK executor transports the gripper to the calibrated bowl. No fruit entity handle is read between grasp planning and release. This shared executor is important: it lets the same action path be evaluated after perception changes without quietly changing the task interface.

4 Perturbation Protocol

The frozen benchmark uses three fruits, conditions listed in Table 1, and seeds $\{11, 23, 37, 53, 71\}$. Every condition preserves the task, fruit set, success predicate, and

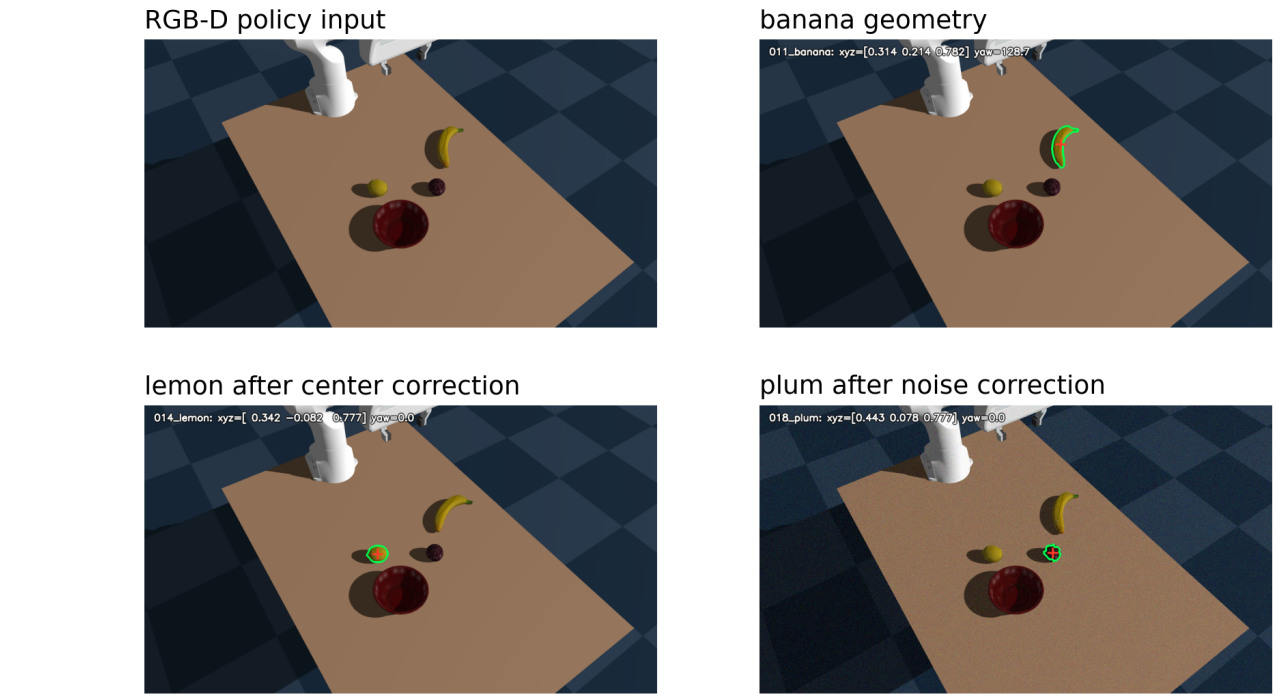


Figure 1: Representative archived evidence. The first two panels are the nominal banana policy input and overlay; the last two are complete post-fix overlays for the formerly failing lemon pose-shift and plum pixel-noise episodes. The frames are direct Genesis outputs, not explanatory renders.

seed protocol while changing one causal axis. The ten runtime conditions contain 150 episodes. Appearance requires Genesis scene rebuild because colors and camera parameters are baked at construction, so it is run as a separate 15-episode matrix. We report conditions separately and do not pool them into a single robustness rate.

RGB and depth perturbations are independent: RGB noise and blur leave depth unchanged, while depth noise and dropout leave RGB unchanged. The camera-view condition moves the camera and supplies its resulting public calibration; it does not evaluate a deliberately stale calibration. Low friction scales the Franka, fruits, and table together because Genesis combines contact-pair friction using the maximum coefficient. Heavy payload scales only the requested fruit and restores all other masses before each episode.

Every reported cell carries an explicit epistemic label so that a perturbation is never mistaken for its own solution: a frozen baseline rate, a matched ablation, a measurement-only diagnostic, or a rejected candidate. Rejected and partial results are retained in the repository rather than discarded, and diagnostics that the episode schema cannot resolve are reported as unresolved rather than assigned a mechanism.

5 Failure-Led Corrections

The first integration smoke matrix intentionally retained failures. Under seed 23, pixel noise fragmented the dark plum mask and produced no surviving candidate. Under pose shift, the lemon candidate existed but the grasp pushed the fruit aside. A moderate occlusion smoke run localized all three fruits, with XY errors of 8.60, 8.84, and 11.57 mm, but only banana completed. These records show why perception error and task outcome cannot be substituted for one another.

Two sensor-only changes were then made. First, plum segmentation receives a 5×5 median pre-filter and requires a component area of at least 500 pixels, major span below 0.09 m, and minor-to-major span ratio at least 0.60. Second, round-fruit center correction replaces the visible-surface median with the calibrated center-height ray intersection described in Section 3. On the identical seed-23 appearance setup, a depth-grounded foreground mask also prevents a recolored table from merging into the yellow fruit component. The retained pre-fix component had 405,312 pixels and 133/155 mm XY errors; the post-fix cells complete with 10.01, 0.79, and 3.28 mm errors.

The corrections were selected from retained failures and then frozen before the five-seed matrix. They do not add simulator labels, object locations, or instance masks. Their role is to repair observable sensing geometry, not to claim broad domain generalization.

6 Experimental Results

6.1 Main benchmark

Figure 2 reports task completion and median evaluator-only XY error by condition. The runtime matrix com-

Table 1: Frozen perturbation contract. Values are the requested ranges; the sampled per-episode values are also written to JSON.

Condition	Changed axis	Operational perturbation	Matrix handling
Nominal	none	unchanged scene and observation	runtime, control
Pose shift	object state	position jitter ± 0.03 m; yaw jitter $\pm 30^\circ$	runtime
Pixel noise	RGB	Gaussian noise $\sigma = 12/255$ in the recorded policy image	runtime
Image blur	RGB	Gaussian blur with a 9×9 kernel	runtime
Depth noise	depth	metric Gaussian noise, $\sigma = 3$ mm	runtime
Depth dropout	depth	10% of valid depth pixels set missing	runtime
Occlusion	visibility	frozen fruit-specific edge strips retaining 77.9–83.5% of color components	runtime
Camera-view shift	viewpoint	camera position and look-at jitter ± 0.015 m; resulting calibration supplied	runtime
Appearance shift	scene appearance	table RGB jitter ± 0.15 and constrained per-class object recoloring	rebuilt scene
Low friction	contact	all relevant contact surfaces scaled to $[0.55, 0.75]$ of nominal	runtime
Heavy payload	dynamics	requested fruit mass scaled to $[1.30, 1.70]$; distractors reset to nominal	runtime

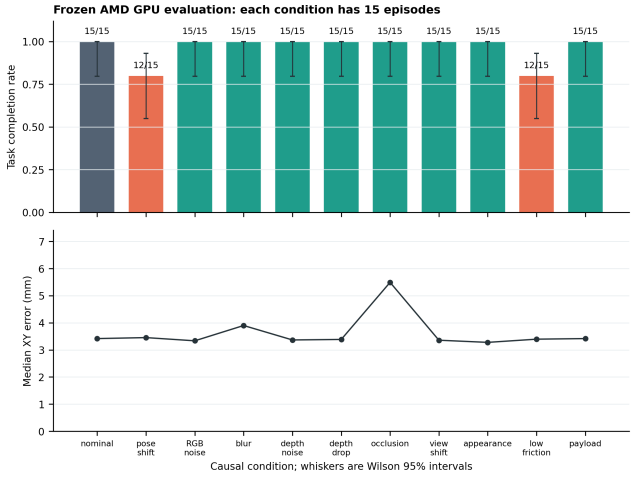


Figure 2: Per-condition results from the frozen baseline matrices. Bars show completion with Wilson 95% intervals; the lower panel shows median initial XY localization error. Appearance is the separate rebuilt-scene condition.

pletes 144/150 episodes. Nominal, pixel noise, blur, depth noise, depth dropout, occlusion, camera-view shift, and heavy payload each complete 15/15. Pose shift and low friction complete 12/15. The appearance matrix completes 15/15. Because 15 trials per cell make the normal approximation unreliable near the boundary, we report Wilson score intervals[12]: 15 completions give 0.796–1.000, while 12 completions give 0.548–0.930.

The failures are concentrated rather than diffuse: all six are lemon `task_failed` outcomes. Pose-shift failures occur at seeds 11, 23, and 71; low-friction failures occur at seeds 23, 37, and 71. Their localization errors are 0.41–0.70 mm. The correct conclusion is therefore that the policy continued to perceive the lemon accurately under these draws, while the contact-and-execution path failed. The available episode schema does not provide phase telemetry sufficient to label these events as slip, drop, or closure loss, so we retain the broad task-failure label.

Table 2: Frozen completion counts and median localization error.

Condition	Done	Median XY (mm)
Nominal	15/15	3.42
Pose shift	12/15	3.45
Pixel noise	15/15	3.34
Image blur	15/15	3.90
Depth noise	15/15	3.37
Depth dropout	15/15	3.39
Occlusion	15/15	5.49
Camera-view shift	15/15	3.36
Appearance shift	15/15	3.28
Low friction	12/15	3.40
Heavy payload	15/15	3.42

6.2 Controller ablations are not perception fixes

To test whether the remaining physical failures could be repaired by a simple force-and-speed guard, we ran a matched five-seed ablation on the low-friction and heavy-payload cells. The guard increases grasp force by 20% and reduces the maximum grasped-motion increment from 0.006 to 0.004 rad per 100 Hz step. As shown in Figure 3, baseline completes 27/30 physical cells, whereas the guard completes 20/30. In particular, lemon low-friction completion falls from 2/5 to 1/5 and plum heavy-payload completion falls from 5/5 to 0/5. A separate 160-step low-force contact guard also fails its targeted lemon low-friction seed-23 cell. These are negative ablations, not mitigations, and are excluded from the baseline rates.

We then instrumented the gripper with finger position and velocity telemetry. Incremental closure declares contact only after command-tracking lag and low velocity remain stable; the resulting event can switch force directly or ramp through 0.25, 0.50, and 0.75 of nominal. After lift, a fresh RGB-D observation searches for target-colour points above a minimum height near the estimate. Failed checks trigger sensor-only re-localization and a bounded retry. The action path receives neither evaluator truth nor the perturbation name.

The follow-up is a controller-selection study, not a replacement benchmark. Pure position hold, direct and adaptive force, force-settle position hold, and ramped-force candidates all failed a fixed five-seed lemon pose gate. The final five-attempt candidate combines event ramp, baseline force, event direct, baseline force with

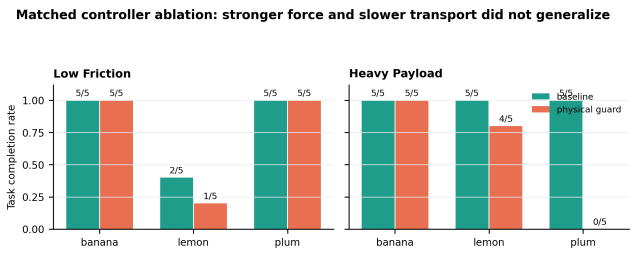


Figure 3: Matched physical controller ablation. The stronger-force, slower-transport guard regresses the physical subset and is not promoted into the primary policy.

a +3 mm XY offset, and event ramp with a −3 mm offset. It completes 4/5 fixed cells: seeds 11, 23, 53, and 71 pass, while seed 37 fails after five attempts. Because the prerequisite gate is not 5/5, the candidate is not promoted and no new 150+15 matrix is run.

The verifier also exposes a measurement limitation. The targeted seed-11 episode passes the independent post-episode evaluator after five attempts, but all five internal lift checks observe zero lifted target pixels and request a retry. The robot hand can occlude a held fruit. Therefore RGB-D lift verification is an auxiliary recovery signal, not ground-truth success.

6.3 Runtime and evidence accounting

The 150 runtime episodes require 1,728.5 s total wall time on the observed environment, with a 10.93 s median and 9.79–15.56 s range per episode. The rebuilt appearance matrix requires 184.5 s total, with an 11.15 s median. Wall-time numbers include simulation and rendering as recorded by the evaluator; they are not a real-time control-rate claim.

Each successful evidence episode can include a policy input image, detection overlay, eight stage frames, an H.264 video, and an `episode.json`. Batch rows additionally store requested ranges, sampled offsets and ratios, estimated pose, evaluator-only initial truth, localization error, task outcome, controller parameters, and wall time. The archived SHA-256 manifest covers the expanded batch. This structure is designed to support an audit of the chain from perturbation to observation to action to outcome.

7 Discussion

The central empirical result is a boundary diagnosis. Visual perturbations that could corrupt the component mask or the RGB-D geometry are tolerated in the tested range after two targeted corrections. Occlusion increases median localization error from 3.42 to 5.49 mm but does not reduce completion in this matrix. Viewpoint perturbation does not cause a regression when the moved camera's calibration is consumed. Appearance shift exposes the importance of depth-grounded foreground selection: color alone is insufficient when the support plane shares the target hue.

The remaining failures are a different class. A sub-millimeter initial XY error does not guarantee a successful grasp of a small round fruit. This is a useful negative result because a superficially plausible physical guard makes the matched result worse and introduces a plum heavy-payload regression. The new follow-up now records finger separation, velocity, command tracking error, contact events, force branches, stage state, and lift decisions. It improves attribution and repairs individual cells, but the changing failed seed across 4/5 gates shows that hand-selected force logic has not produced stable robustness. Better visual coverage or tactile sensing is needed before a recovery policy can be trained across observed contact outcomes.

The design also illustrates a practical complementarity with current VLA research. Learned systems can offer broader semantic and cross-view generalization[6, 5, 4]; a small calibrated policy offers an interpretable control condition against which that extra capability can be measured. A useful future benchmark would place both policies behind the same information boundary, perturbation draws, success predicate, and failure telemetry.

8 Limitations and Reproducibility

The detector is class-specific and receives the requested class. The bowl is a fixed calibrated fixture rather than a visually localized object. Appearance variation is synthetic and constrained by class priors. The benchmark uses five seeds per condition and one simulated scene, so intervals describe these episodes rather than a population-level deployment guarantee. No learned detector, SmolVLA result, end-to-end learned policy, real robot result, or sim-to-real transfer claim is made.

The post-lift RGB-D verifier is conservative under hand occlusion. A zero-pixel decision can be a false negative and must not be interpreted as evaluator truth. The controller follow-up is a small candidate-selection gate and is not pooled with the frozen baseline. Matched-seed Oracle execution, tactile sensing, additional camera coverage, and a learned recovery policy remain future work.

The upstream simulator is pinned at commit `8cd9dd28396c8090b2747948a05ba14a0f6136a8`. The repository provides the patch under `simulation_patch/franka_fruit_pick/`, the exact runtime and appearance commands under `docs/REPRODUCIBILITY.md`, unit tests for the observation and perturbation contract, and the archived evidence under `runs/vision-guided-rocm-20260805/` and `runs/vision-guided-rocm-20260806/`. The sensor/controller follow-up, including its final 4/5 gate, representative rejected profiles, and targeted seed-11 video, is archived separately under `runs/vision-guided-rocm-20260806-followup/`. A deterministic CPU reference path is included for schema validation, but it is not counted as AMD GPU evidence. The real claim is tied to archived `genesis-amdgpu` JSON and its environment receipt.

9 Conclusion

We presented an auditable RGB-D manipulation policy and a frozen causal robustness protocol executed on AMD Radeon/ROCm. The policy removes simulator object state from the action path, converts calibrated observations into a grasp pose, and retains enough evidence to distinguish perception from execution. Across 165 baseline cells, the tested visual and geometric perturbations complete reliably, while six lemon failures reveal a contact boundary despite sub-millimeter localization. Sensor-level contact and lift telemetry makes that boundary observable, but a final 4/5 controller gate is not promoted. Negative and partial ablations show why mitigation claims require matched evaluation. The resulting system is a compact, reproducible reference for future learned perception, recovery, and failure-monitoring baselines.

Author Contributions

Ziyu Zhou led the project: system design, the sensor-only RGB-D policy, the perturbation protocol, execution on the AMD Radeon/ROCm environment, the evidence pipeline, and the manuscript. Yunjia Zhang contributed to evaluation review, reproducibility testing of the documented commands, and the reproducibility materials. Liqing Sang contributed to evidence archiving and checksum verification, and to the demonstration video. All authors reviewed and approved the final report and take responsibility for the reported results.

Artifact and Code Availability

The code, experiment contracts, JSON records, checksum manifests, frame/video evidence, and compilation instructions are included in the accompanying repository. The paper figures are generated by `paper/make_figures.py` directly from the archived JSON and image files.

References

[1] Genesis Team, "Genesis: A generative and universal physics engine for robotics and beyond," project documentation and software release, 2024–2026.

[2] B. Calli et al., "The YCB object and model set: Toward common benchmarks for manipulation research," in *Proc. IEEE International Conference on Advanced Robotics*, 2015.

[3] J. Tobin et al., "Domain randomization for transferring deep neural networks from simulation to the real world," in *Proc. IEEE/RSJ IROS*, 2017.

[4] Q. Wang et al., "RayViT: Ray-conditioned visual representations for viewpoint-robust imitation learning," arXiv:2607.29622, 2026.

[5] T. Zhang et al., "OC-VLA++: Monocular geometry-guided cross-view consistency for viewpoint-robust robotic manipulation," arXiv:2608.01066, 2026.

[6] P. Li et al., "BridgeVLA++: A data-efficient, generalizable, and memory-augmented vision-language-action framework for 3D manipulation," arXiv:2608.05042, 2026.

[7] S. Hegde and J. Y. B. Katta, "GUARD: Grounding uncertainty and ablation-based risk detection for diffusion-based VLAs," arXiv:2608.04510, 2026.

[8] H. Rajaprakash et al., "SAFECAST: Robust failure detection for VLA policies with contrast-set training and calibration," arXiv:2608.04246, 2026.

[9] S. Wang et al., "Suppression sticks, locality is fragile: A closed-loop target-and-control audit of task-vector negation in VLA policies," arXiv:2608.04692, 2026.

[10] BWM Team, "BWM: A low-cost high-fidelity world simulator for robot learning," arXiv:2607.29302, 2026.

[11] Franka Robotics, "Franka Emika Panda research robot," product and control documentation.

[12] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, 1927.